Name _____ Surname _____ Date _____

Test: Measurement and laboratory skills

Version A

Duration: 40 minutes. Total: 60 marks. A calculator is allowed; there is no data sheet. Write your answers on this sheet. Give the unit in every answer that needs one. In the multiple-choice questions (1, 14 and 17) only one answer is correct: write its letter in the box. A wrong answer scores the same as no answer. In questions 15 to 20 show your working.

> **Safety net.** The questions marked ⛑ are worth 36 marks out of 60, that is 60%: if you answer all of them correctly, you pass.

Part A

1. ⛑ Which of these is a derived quantity? **[2 marks]**

A) length **B)** mass **C)** time **D)** density Answer: ☐

2. ⛑ Write the SI unit of:

(a) volume _____ **[1 mark]** (b) density _____ **[1 mark]**

3. ⛑ Convert:

(a) 0.75 kg = _____ g **[1 mark]** (b) 36 mm = _____ m **[1 mark]**

4. ⛑ Convert 6.4×10^4 mm to m. Give the answer in standard form. **[2 marks]**

Answer: _____ m

5. ⛑ Write in standard form:

(a) 0.000 074 s = _____ s **[1 mark]** (b) 3 600 000 m = _____ m **[1 mark]**

6. ⛑ Write the number of significant figures of each measurement:

(a) 0.00240 s _____ **[1 mark]** (b) 5.0×10^3 m _____ **[1 mark]**

7. ⛑ Round each number to the number of significant figures shown:

(a) 3.14159 s to 3 significant figures: _____ s **[1 mark]**

(b) 0.01996 m to 2 significant figures: _____ m **[1 mark]**

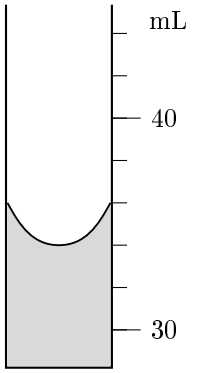
8. ⛑ Write each number in standard form, with the number of significant figures shown:

(a) 25 480 m, 2 significant figures: _____ m **[1 mark]**

(b) 0.000 3047 s, 3 significant figures: _____ s **[1 mark]**

9. ⛑ The diagram shows the water in a measuring cylinder. Read the volume of water. **[2 marks]**

Answer: _____ mL



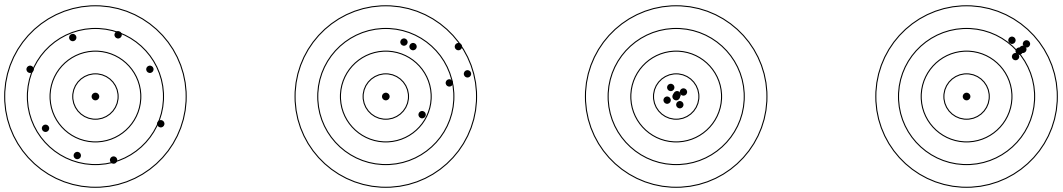
10. ⛑ What is the uncertainty of one reading on this cylinder? **[2 marks]**

Answer: ± _____ mL

11. ⛑ With nothing on it, a digital balance reads 0.6 g. A stone placed on the balance makes it read 24.3 g. Write the correct mass of the stone. **[2 marks]**

Answer: _____ g

12. Four students throw darts. Each dart board stands for repeated measurements of one quantity, and the bullseye is the true value.



1 2 3 4

(a) For each board, write yes or no. **[4 marks]**

Board	Accurate?	Precise?
1		
2		
3		
4		

(b) Write the numbers of the boards that show: **[2 marks]**

a systematic error: _____ a large random error: _____

(c) A student always starts the stopwatch 0.3 s late and stops it exactly on time. Which board shows how his readings look? Is his error random or systematic? **[2 marks]**

Board: _____ Error: _____

(d) Complete the sentence. **[2 marks]**

Taking the mean of many readings reduces _____ error, but not _____ error.

13. In the pendulum lab you change the length of the string and measure the time for 20 swings. Write one quantity that must be kept the same (a controlled variable). **[2 marks]**

Answer: _____

14. In Google Sheets, cells B2 and C2 hold the times for 20 swings of the same pendulum, $t_1 = 18.6$ s and $t_2 = 18.4$ s. Cell D2 is empty. Which formula, typed in E2, gives the mean time? **[2 marks]**

A) `=AVERAGE(B2:D2)` **B)** `=SUM(B2:D2)` **C)** `=B2+C2+D2/3` **D)** `=MAX(B2:D2)`

Answer: ☐

Part B

15. A student times a toy car rolling down a ramp five times: 3.42 s, 3.38 s, 3.40 s, 4.01 s and 3.41 s. The 4.01 s result is anomalous. Calculate the mean of the other four readings. Give your answer to 3 significant figures. **[4 marks]**

Answer: _____ s

16. In the pendulum lab, a group times 20 swings and gets 33.2 s. The uncertainty in starting and stopping the stopwatch is ±0.2 s. Calculate the time for one swing, with its uncertainty. **[4 marks]**

Answer: $T =$ _____ s ± _____ s

17. A pendulum has a period of about 1.6 s, and the uncertainty in the total time is ±0.2 s. What is the smallest number of swings you must time so that this uncertainty is less than 1% of the total time? **[2 marks for the answer, 2 marks for the reason]**

A) 10 swings **B)** 12 swings **C)** 13 swings **D)** 20 swings Answer: ☐

Why? _____

18. In the density lab a student reads 25 mL before and 42 mL after a stone is put in. Each reading has an uncertainty of half a division (0.5 mL). Calculate the relative uncertainty of the volume of the stone. **[4 marks]**

Answer: _____ %

19. The same stone has a mass $m = 52.7 \pm 0.1$ g and a volume $V = 17 \pm 1$ cm^3.

(a) Calculate the density $\rho = m/V$. Give the unit. **[2 marks]**

Answer: _____

(b) Calculate the relative uncertainty of the density. **[2 marks]**

Answer: _____ %

20. Use your answers to question 19. Calculate the absolute uncertainty of the density and write the density of the stone in the form $\rho \pm \Delta\rho$. Round the uncertainty to 1 significant figure and give ρ to the same number of decimal places. Give the unit. **[4 marks]**

Answer: $\rho =$ _____ ± _____ _____